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Senescence of human visual-vestibular interactions: smooth pursuit, optokinetic, and vestibular control of eye movements with aging

Received: 4 February 1993 / Accepted: 1 October 1993

Abstract Natural aging entails progressive deterioration in a variety of biological systems. This study focuses on visual and vestibular influences on human eye movements as a function of aging. Eye movements were recorded (search-coil technique) during visual, vestibular, and combined stimuli in subjects across a broad range of ages (18–89 years). Two types of visual following were assessed: smooth pursuit (SP) of a small discrete target, and optokinetic (OKR) following of a large-field striped image. The vestibulo-ocular reflex (VOR) was studied during head rotation in darkness. Visual-vestibular interactions were recorded during rotation in two ways: when the optokinetic scene was earth-fixed, resulting in visual enhancement of the VOR (VVOR), and when the visual image was head-fixed, allowing visual suppression of the VOR (VSVOR). Stimuli consisted of horizontal sinusoidal oscillations over the frequency range 0.025–4 Hz. Trials were analyzed to yield response gain (peak horizontal eye/stimulus velocities) and phase (asynchrony, in degrees, between eye and stimulus velocity signals). VOR gain in young subjects was greatest (near 0.9) at 2.5–4 Hz but declined steadily with decreasing frequency, while phase hovered near zero until 0.1 Hz and then developed a progressively increasing lead. Effects of advancing age were small, given the modest head velocities presented, and were most noticeable as an increase in phase lead and decline in gain at the lowest frequencies (≤ 0.1 Hz). The two forms of visual following and all conditions of visual-vestibular interactions displayed more prominent age-dependent changes. OKR and SP response characteristics (0.25–4 Hz) closely resembled each other. Gain was greatest at 0.25 Hz, while phase was near 0°. As frequency increased, gain declined while phase lag rose. However, both gain and phase lag tended to be slightly greater for OKR than for SP responses. Both SP and OKR

response properties deteriorated progressively with increasing age, as witnessed by a progressive decline in gain and increase in phase lag, even at modest frequencies (e.g., 0.25–1.0 Hz). VVOR responses were generally closer to the ideal of 1.0 in gain and 0° in phase than either the VOR or visual following alone. A subtle but significant age-dependent decline in VVOR performance occurred at the lowest frequencies. VSVOR response characteristics were close to those of the VOR and VVOR at 4 Hz, where visual influences on eye movements are generally inconsequential. As frequency declined, visual suppression became more robust and gain dropped. The SP stimulus seemed surprisingly more effective than the OK scene in suppressing the VOR, but this effect is predicted by a linear model of visual-vestibular interactions. As age increased, visual influences on the VOR became progressively weaker, in concert with deterioration of visual following. The subjective sensation of circular vection (CV), a psychophysical measure of VVI, was assessed during optokinetic stimulation at 0.025 Hz. Interestingly, the likelihood and intensity of CV increased with aging, suggesting that visual inputs to the perception of self-motion are enhanced in the elderly. This may represent a form of visual compensation for age-dependent loss of vestibular self-rotation cues. In brief, the VOR, visual following, and their interactions display specific changes in response properties as a function of natural aging. The modifications may be interpreted as age-dependent deteriorations in the performance of systems underlying the control of human eye movements.

Key words Smooth pursuit · Optokinetic
Visual-vestibular interactions · Aging · Human

Introduction

Aging entails deteriorations in a variety of biological systems, including those that control equilibrium and orientation in space. Interest in this area is motivated by

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an important public health concern; falls and their consequences (including death) progressively increase with aging (Radebaugh et al. 1985). A fall is a disastrous failure of equilibrium control which presumably reflects errors made within underlying neural subsystems. Unfortunately, visual and vestibular control of eye movements have received relatively limited attention, despite their important role in maintaining spatial orientation. This study addresses this area. In addition, several concerns relevant to the different types of visually driven eye movements (i.e., optokinetic and smooth pursuit) are included.

Natural behavior typically includes head movements in the presence of a rich visual world. Several mechanisms exist by which motion of the head and/or visual elements produce eye movements that maintain binocular fixation on targets of interest. Confining discussion to angular motion, head rotation activates the semicircular canals, which generate compensatory eye movements that counteract head motion through the vestibulo-ocular reflex (VOR). Visual inputs also drive eye movements, and through two major subsystems. First, the smooth pursuit system (SP) allows primarily foveate species to track discrete visual targets. Second, the optokinetic (OK) system (or "reflex," OKR) allows all known species with mobile eyes to follow large-field visual scenes. However, the OKR in foveate animals differs from that in others, since there is nothing to prevent the SP system from joining in the response to moving visual elements in the field (Lisberger et al. 1981b; Robinson, 1981). The primate OKR is usually described as a two-component process: a fast component that resembles SP and is driven by central retinal inputs, and a slow component that is driven by widespread retinal elements (Cohen et al. 1977; Dichgans and Brandt 1978; Lisberger et al. 1981b; Robinson 1981; Paige 1983a). Although the SP and OK systems have been extensively studied, few have assessed both in the same subjects (Yee et al. 1983; Yasui and Young 1984; Van den Berg and Collewijn 1986), and none have compared the two across ages.

The VOR and visual following systems typically operate in concert during natural behavior. The resulting visual-vestibular interactions (VVI) entail direct moment-to-moment effects of vestibular and visual influences on eye movements. VVI is commonly studied in two ways, during head rotation in the presence of: (1) earth-fixed images, thereby allowing visual inputs to enhance vestibular-driven ocular responses (VVOR); and (2) head-fixed images, thereby allowing visual inputs to suppress vestibular-driven eye movements (VSVOR). Although VVI has been studied in the past (Paige 1983a; Barnes 1988; Jell et al. 1988; Kasteel-van Linge and Maas 1990), direct comparisons between SP and OK stimuli during the two types of VVI, and across ages, are lacking.

The goal of this study was to investigate the human VOR and visual following systems (SP and OKR) in isolation and during synergistic (VVOR) and antagonis-

tic (VSVOR) visual-vestibular interactions. Normal human subjects over a broad range of ages were studied in order to characterize these ocular responses as a function of natural aging. A complete report of VOR findings has been published previously (Paige 1992). A subset of the results are included here; namely, VOR responses to rotations whose characteristics matched those of visual and VVI stimuli. Visually driven responses to both SP and OK stimuli are then presented, along with relevant comparisons between their response dynamics. Finally, interactions between the VOR and both the SP and OK systems are detailed. In general, stimulus characteristics (frequencies and amplitudes) across all six conditions (VOR, SP, OKR, and their interactions) were matched to facilitate comparisons and to assess the linearity of vestibular and visual influences during combined stimulation.

Materials and methods

Subjects

Eighty-one normal human volunteers between 18 and 89 years of age were studied. All subjects were non-institutionalized and self-sufficient. None had a history of past or present neurologic, ophthalmologic, otologic, systemic, or traumatic disease that potentially could have affected results. All showed normal age-corrected function on a clinical examination of neurologic, ophthalmologic, and otologic function (performed by the author), with particular attention toward visual acuity (20/40 or better in both eyes, corrected), stereopsis, visual fields, binocular oculomotor control, binaural hearing (pure tone audiometry), and bilateral vestibular function (postural and caloric tests). Eight subjects were eliminated due to the development of untoward symptoms during the study. Data are reported from the remaining 73. Additional details on the same subjects are available in a more comprehensive report on the VOR alone (Paige 1992).

Eye movement recording

Horizontal and vertical eye movements were recorded unilaterally using the magnetic search-coil technique (Robinson 1963; Collewijn et al. 1975), as described previously (Paige and Sargent 1991). Horizontal and vertical eye position was recorded with a resolution of $<0.05^\circ$. Calibration of eye position was obtained by instructing subjects to fixate a discrete projected target (0.3° diam.) at 0° , $+10^\circ$, and -10° . Vertical eye movements were monitored only to assure that vertical gaze was within 10° of the horizontal plane.

Stimulus characteristics

Details of the rotatory stimulus were provided previously (Paige and Sargent 1991). Briefly, subjects sat in a custom non-metallic chair with their heads secured in a six point head fixation device. The chair was rigidly mounted on a servo-controlled rate table (Contraves-Goerz 824), and subjects were rotated sinusoidally at seven frequencies ranging from 0.025 to 4.0 Hz. Peak velocity was held constant at $50^\circ/\text{s}$, except at 2.5 and 4.0 Hz, where torque limitations required reducing peak velocity to $25^\circ/\text{s}$ and $10^\circ/\text{s}$, respectively. Head rotation faithfully followed the chair tachometer signal at all but the highest frequency. Even at 4 Hz, gain and phase measurements (see Data collection and analysis) were in error by at most 0.03 and 3° , respectively (Paige and Sargent 1991). For some early experiments, mechanical shortcomings of a

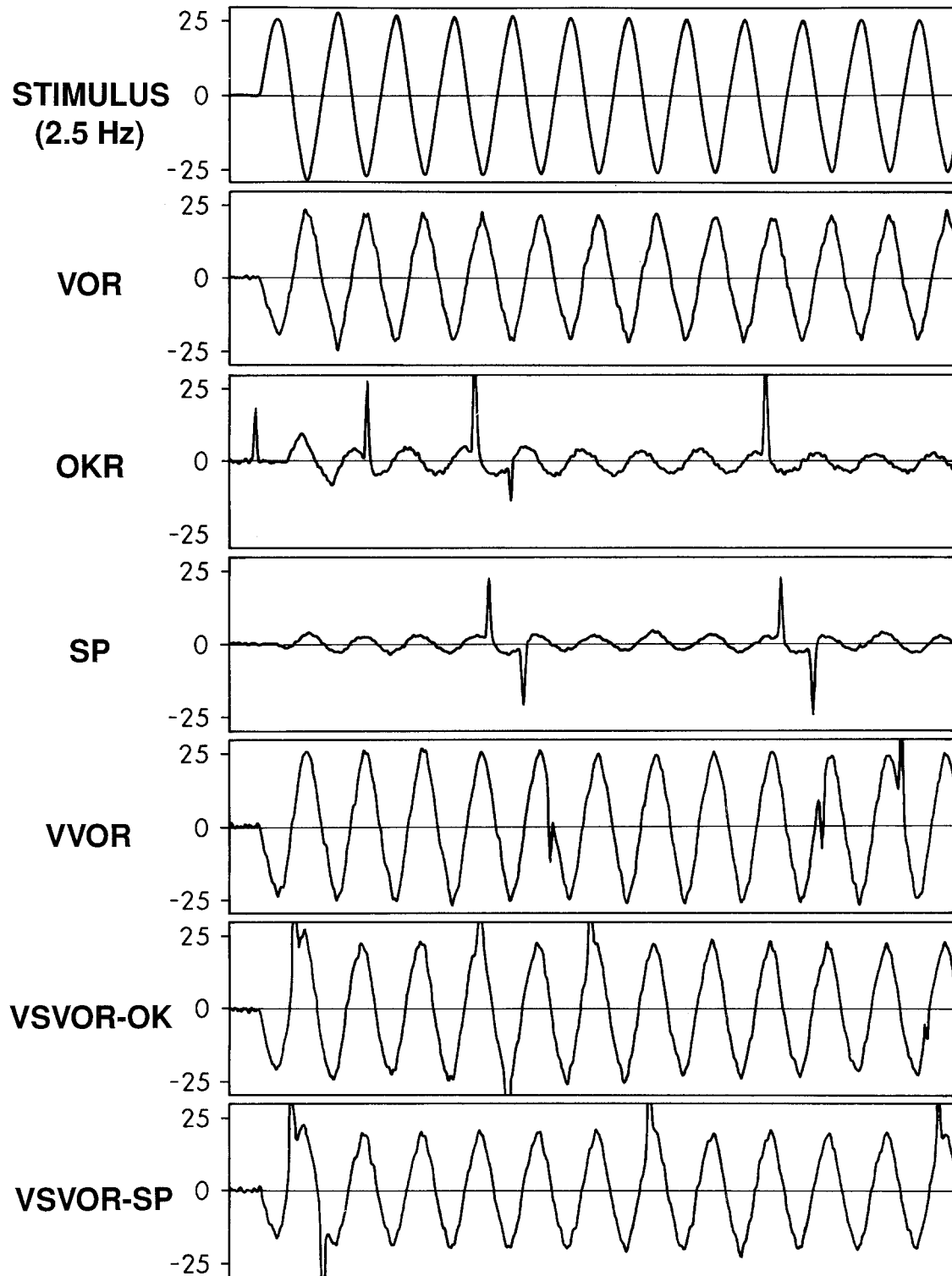


Fig. 1 Recordings of eye velocity responses to stimulus oscillation at 2.5 Hz under all six conditions studied (VOR, OKR, SP, VVOR, VSVOR-OK, VSVOR-SP). All traces are from one subject (53 years). The stimulus (*top trace*) applies accurately to all responses except SP, in which the relationship between the stimu-

lus and response is shifted 90° (the onset is, however, accurate). Spike-like transients in response traces correspond to small saccades and blinks. They are clipped at the limits of each trace's display field

Table 1. Response gain and phase for all experimental conditions

Ampli- tude	Fre- quency	Young					Middle-aged					Elderly				
		<i>n</i>	Gain	SD	Phase	SD	<i>n</i>	Gain	SD	Phase	SD	<i>n</i>	Gain	SD	Phase	SD
VOR																
50	0.025	30	0.53	0.13	14.3	5.8	23	0.49	0.15	17.2*	5.8	20	0.46*	0.12	24.6&	12.7
50	0.10	27	0.64	0.11	2.1*	3.1	22	0.57	0.16	4.7	4.3	20	0.57	0.15	5.5*	6.6
50	0.25	29	0.69	0.11	0.0	2.1	23	0.69	0.12	0.0	4.2	19	0.66	0.14	0.4	4.9
50	0.50	24	0.71	0.12	−0.2	2.4	23	0.74	0.16	−0.3	2.7	20	0.80*	0.12	−0.4	3.0
50	1.0	29	0.79*	0.10	2.6	2.8	23	0.85	0.09	1.0	3.3	20	0.86*	0.10	0.6*	2.0
25	2.5	27	0.90*	0.09	2.1	2.2	23	0.85	0.11	1.1	5.8	20	0.86	0.10	0.6	3.4
10	4.0	24	0.87	0.12	2.1	4.5	22	0.81	0.13	2.5	5.6	19	0.82	0.13	0.7	5.2
OKR																
50	0.025	28	0.76	0.17	0.5	3.8	17	0.66	0.18	−0.8*	5.2	17	0.61 #	0.14	−4.8&	4.5
50	0.10	26	0.79*	0.16	0.4*	7.3	15	0.68	0.15	−5.0	6.0	16	0.61 #	0.18	−5.5 #	5.9
50	0.25	28	0.84	0.10	0.4&	2.5	20	0.81*	0.10	−3.5	3.8	19	0.69&	0.18	−6.7 #	12.5
50	1.0	28	0.69*	0.15	−10.7&	9.7	20	0.59 #	0.16	−29.6	21.6	19	0.43&	0.17	−35.6&	28.4
25	2.5	11	0.18	0.11	−63.6*	18.8	17	0.17 #	0.10	−92.3	34.9	13	0.09 #	0.04	−85.8*	21.4
10	4.0	11	0.08	0.04	−136.6&	20.2	16	0.08	0.04	−177.5	31.7	12	0.06	0.02	−190.3 #	50.6
SP																
50	0.25	27	0.84	0.09	2.5*	2.2	20	0.79 #	0.11	1.1	2.1	19	0.60&	0.22	2.2	5.3
50	1.0	28	0.60	0.14	−9.2&	4.0	22	0.53 #	0.14	−16.7	9.2	18	0.37&	0.16	−14.6&	5.9
25	2.5	24	0.20	0.10	−46.0	13.6	18	0.19*	0.10	−51.9	17.5	17	0.11 #	0.06	−50.0	14.9
10	4.0	22	0.09	0.04	−104.7 #	17.1	19	0.07	0.04	−134.9	39.1	15	0.06 #	0.03	−126.7*	41.1
VVOR																
50	0.025	28	0.98	0.04	1.0	1.3	23	0.95*	0.09	2.5	4.4	19	0.88&	0.09	3.2&	2.4
50	0.10	29	1.01	0.03	0.3	0.4	23	1.00*	0.04	0.3*	0.5	19	0.96&	0.06	0.8*	0.9
50	0.25	29	1.01	0.04	0.0*	0.7	23	1.01 #	0.04	−0.5	0.7	19	0.96 #	0.06	0.1	1.8
50	1.0	29	1.02	0.06	−0.2 #	0.8	23	1.02	0.06	−1.2	1.3	19	1.01	0.06	−0.6	1.3
25	2.5	26	0.95	0.07	0.0	1.6	23	0.92	0.09	−1.1	2.9	19	0.93	0.09	−0.7	2.5
10	4.0	24	0.90	0.10	1.4	3.2	23	0.85	0.11	2.9	5.3	18	0.88	0.11	0.4	3.2
VSVOR-OK																
50	0.25	27	0.05	0.04	30.9	49.5	15	0.07	0.04	29.1	29.2	17	0.12 #	0.11	18.5	9.8
50	1.0	28	0.27	0.10	37.6	22.7	15	0.31	0.10	37.2*	14.2	17	0.39 #	0.15	26.1	12.1
25	2.5	10	0.85*	0.11	13.5	6.5	15	0.73	0.13	13.9	11.1	12	0.78	0.08	7.5*	5.0
10	4.0	10	0.94	0.14	5.1	3.4	15	0.86	0.13	8.0	9.6	12	0.81*	0.12	3.1	5.1
VSVOR-SP																
50	0.25	27	0.05	0.03	3.3	14.2	16	0.07*	0.03	6.2	12.4	17	0.10&	0.06	13.0	10.1
50	1.0	25	0.23	0.09	22.8	17.2	16	0.26*	0.09	21.4 #	9.1	17	0.35 #	0.14	13.0*	7.5
25	2.5	23	0.74 #	0.11	14.7	7.8	15	0.65	0.06	14.3*	8.2	15	0.70	0.11	8.6 #	4.2
10	4.0	21	0.93 #	0.11	11.4	4.5	15	0.82	0.09	11.6	8.7	13	0.77 #	0.12	9.0	5.4
Statistical comparisons		Young vs middle aged					Middle-aged vs elderly					Elderly vs young				

* $P < 0.05$; # $P < 0.01$; & $P < 0.001$

prototype head-holder limited useful stimulus frequencies to ≤ 1 Hz. During rotations in darkness, subjects were engaged in conversational naming tasks to maintain vigilance and attention. They were also instructed to imagine an earth-stationary structured world directly in front of them.

Optokinetic stimuli were presented using a miniature halogen planetarium bulb inside a cylindrical projection mask. The mask contained 36 vertical equally spaced slits, resulting in a projected image of alternating dark and light stripes, each subtending 5° . The projector was fixed to a velocity servo-controlled motor (Glentek) mounted above the subjects' head. The striped image was projected onto a cylindrical screen 2.4 m in diameter and 1.5 m high, effectively filling the visual field. Stimulus characteristics were the same as those described above for the VOR, except that 0.5 Hz was not tested. However, additional rotations at 1.0 Hz were performed, at 25 and 100°/s peak velocity, in order to test OKR response linearity as a function of stimulus amplitude. For all OKR trials, subjects were instructed to closely follow the stripes passing in front of them.

Smooth pursuit stimuli were presented using a Ne-He laser (Uniphase) projector mounted on the chair just above and behind the head. The laser beam was intercepted by an X-Y galvanometer (General Scanning) and guided onto the projection screen, producing a 0.3° diameter red spot. Horizontal oscillations of the spot were restricted to 0.25–4 Hz, yielding a peak excursion of 31.8° at 0.25 Hz, 50°/s peak target velocity. VVI trials in which the SP target was employed were restricted to VSVOR-SP; the beam was simply fixed relative to the chair, and therefore faithfully followed the head.

VVI was studied when the head was rotated in the presence of an earth-fixed (VVOR) and a head-fixed (VSVOR-OK) OK field. In addition, the VSVOR paradigm was studied with a head-fixed SP target replacing the OK field (VSVOR-SP). For all trials in which visual stimuli were present, subjects were instructed to closely follow the spot or the stripes.

All rotatory stimuli and recordings were controlled by the laboratory's DEC PDP 11-73 computer (see Paige and Sargent 1991). The recording of a complete set of trials required around

35 min. Examples of raw records obtained in response to 2.5 Hz oscillation under all six stimulus conditions (VOR, OKR, SP, VVOR, VSVOR-OK, VSVOR-SP) are shown in Fig. 1 for one subject.

Data collection and analysis

Horizontal head velocity (chair tachometer), OK field velocity, SP target position, and eye position signals during trials were digitally sampled at 200 Hz. For SP trials, target position was digitally differentiated and smoothed (without phase shift) to yield target velocity. Eye position responses were also differentiated to yield eye velocity. Saccades and fast components of nystagmus were then removed using an iterative fitting (least-squares sinusoid) and windowing procedure (Paige and Sargent 1991). The removed portions of the record were replaced by values derived by the least-squares sinusoidal fit for each cycle. Cycles for averaging were then selected, eliminating those with obvious artifact or response distortion (suppression, blinks, squints). Such cycles were easy to identify during analysis. Some trials were eliminated from analysis due to failure to obtain at least two reliable response cycles. This, together with the 1-Hz limit for early experiments, resulted in variability in the number of observations over the data set (see *n* values in Table 1). After cycle selection, both the stimulus and eye velocity signals were averaged over cycles and subjected to Fourier analysis. The fundamentals were used to calculate gain (peak eye velocity divided by peak stimulus velocity) and phase (phase of eye velocity minus phase of stimulus velocity). By convention, phase for trials that entailed head rotation (all but SP and OKR trials) was shifted by 180° such that a perfectly compensatory response was taken as 0°.

Potential effects of aging on response parameters were first sought by splitting the population into three groups (Paige 1992): young (18–44 years, *n* = 30), middle-aged (45–69 years, *n* = 23), and elderly (70–89 years, *n* = 20). Means were then compared across age groups for each trial type. Alternative groupings were employed for specific purposes, as presented below. In all cases male and female subjects were pooled, since separate analysis revealed no significant systematic differences between the sexes in ocular responses to either vestibular or visual stimuli.

The complete data set, including means, SDs, and *n*'s for all trial types (e.g., VOR, SP, etc.) and frequencies are listed in Table 1. The Table also provides statistical comparisons between age groups (student's *t* tests, corrected for unequal *n*'s) for each trial type and frequency. Each of the three age groups is represented by a set of columns. Symbols adjacent to gain and phase values depict statistical significance for specific comparisons between age groups, as identified between brackets at the bottom of the table. For example, the symbols within the "young" set of columns correspond to comparisons between the young and middle-aged groups (see label at the bottom of the table).

Results

Vestibulo-ocular reflex

Mean VOR gain and phase for young, middle, and elderly age groups are presented in Fig. 2 and in Table 1 as a function of frequency, while a raw record during 2.5 Hz oscillation from one subject is included in Fig. 1. The VOR data have been presented separately and more extensively in a previous publication (Paige 1992). A portion of the results are included here because of the direct relevance to VVI and modeling considerations.

VOR gain and phase are similar between age groups (Fig. 2 and Table 1). Notably, response characteristics (both gain and phase) at the highest frequencies are

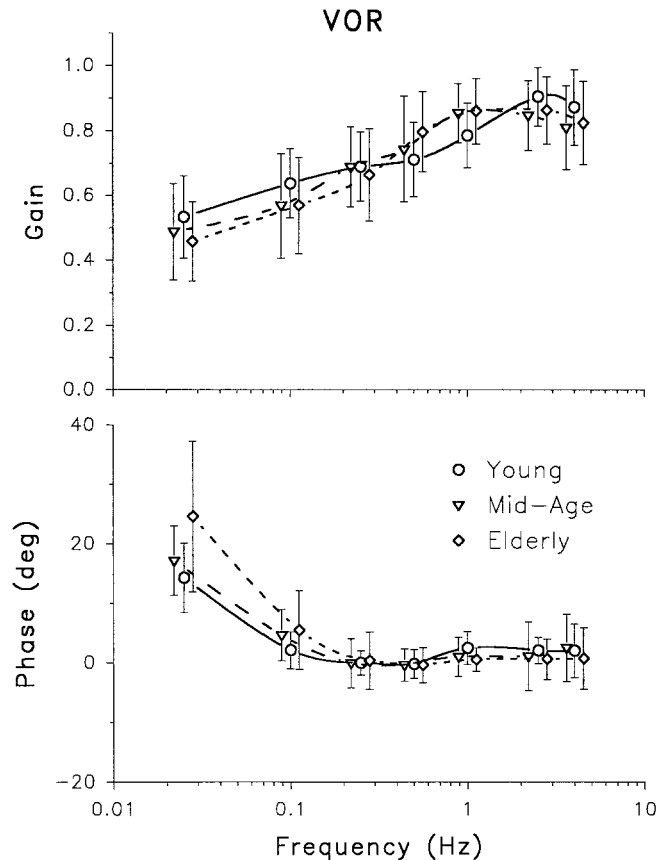


Fig. 2 Mean VOR gain (*top*) and phase (*bottom*) as a function of frequency (peak amplitude $\leq 50^\circ/\text{s}$) for all age groups (see key for symbols). The data for the middle-aged and elderly groups are offset in frequency for graphical clarity. Error bars in this and other figures represent SDs. (data are from Paige 1992)

statistically indistinguishable between groups. In contrast, important differences appear at the lowest two frequencies (0.025 and 0.1 Hz), where phase lead is significantly greater in elderly than in young subjects. This is associated with lower gain in the elderly, but significantly so only at 0.025 Hz. A less striking difference exists between middle-aged and elderly groups at 0.025 Hz, but not between young and middle-aged subjects. Note that variance of phase at low frequencies is greater in the elderly than in the other groups. This represents a true change in the range of values recorded, and not a sporadic influence of outliers.

Visual following

The two types of visually driven eye movements, OKR and SP, are shown for all age groups in Fig. 3 (and Table 1) as mean gains and phases across frequencies, while raw records are included in Fig. 1. OKR and SP responses are initially described separately, followed by a further assessment of their similarities.

Optokinetic

For all age groups, gain is greatest at 0.025–0.25 Hz and declines rapidly with increasing frequency to approach

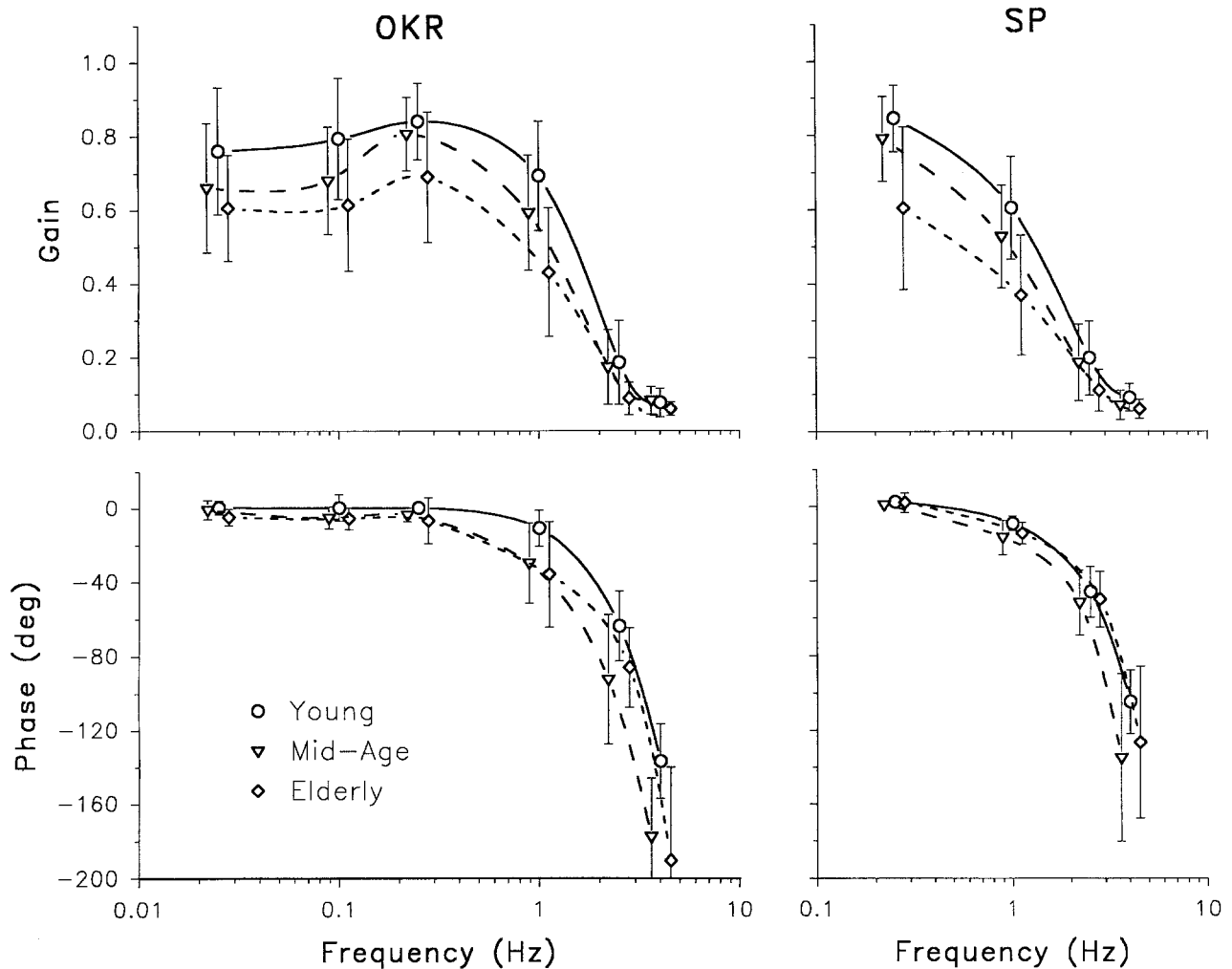


Fig. 3 Mean OKR and SP gain and phase as a function of frequency for all age groups, displayed as in Fig. 2

phase lag over the same range of stimulus amplitudes proves insignificant.

zero at 4 Hz (Fig. 3). Phase lag is near 0° at 0.025–0.25 Hz, but rises progressively as frequency increases above 0.25 Hz.

As age progresses from young to middle-aged to elderly, gain tends to drop while phase lag rises at each frequency, resulting in robust differences between the young and elderly groups (see Table 1 for statistical comparisons). The large variance in phase seen at 4 Hz is due to small gains (and low signal/noise) at this high frequency.

The OKR was also studied at a fixed frequency of 1 Hz, but over a range of amplitudes (25, 50, and $100^\circ/\text{s}$ peak velocity). Results are shown in Fig. 4. In each age group, gain drops roughly monotonically with increasing stimulus amplitude. The effect is nearly parallel between age groups, suggesting that this amplitude-dependent nonlinearity in OKR gain is independent of age. Over all subjects, the drop in OKR gain observed between $25^\circ/\text{s}$ and $100^\circ/\text{s}$ stimuli is statistically significant ($P < 0.001$). In contrast, the more variable changes in

Smooth pursuit

SP response properties resemble those for the OKR (Fig. 3, Table 1). For all age groups, SP gain is greatest at the lowest frequency (0.25 Hz) while phase displays a small lead. As frequency increases, gain drops while phase lag rises dramatically. As age progresses, gain declines at each frequency, while phase lag tends to increase (see Table 1 for statistical comparisons).

As for the OKR, SP tracking was studied at 1 Hz over a $25\text{--}100^\circ/\text{s}$ range of peak velocities (Fig. 4). Within each age group, gain drops steadily with increasing peak target velocity, accompanied by a subtle increase in phase lag. These changes are nearly the same (parallel) between age groups, again implying that the amplitude-dependent changes (nonlinearities) are independent of age. Over all subjects, the changes in response characteristics induced by a fourfold increase in stimulus amplitude are statistically significant ($P < 0.001$ for gain, $P < 0.01$ for phase).

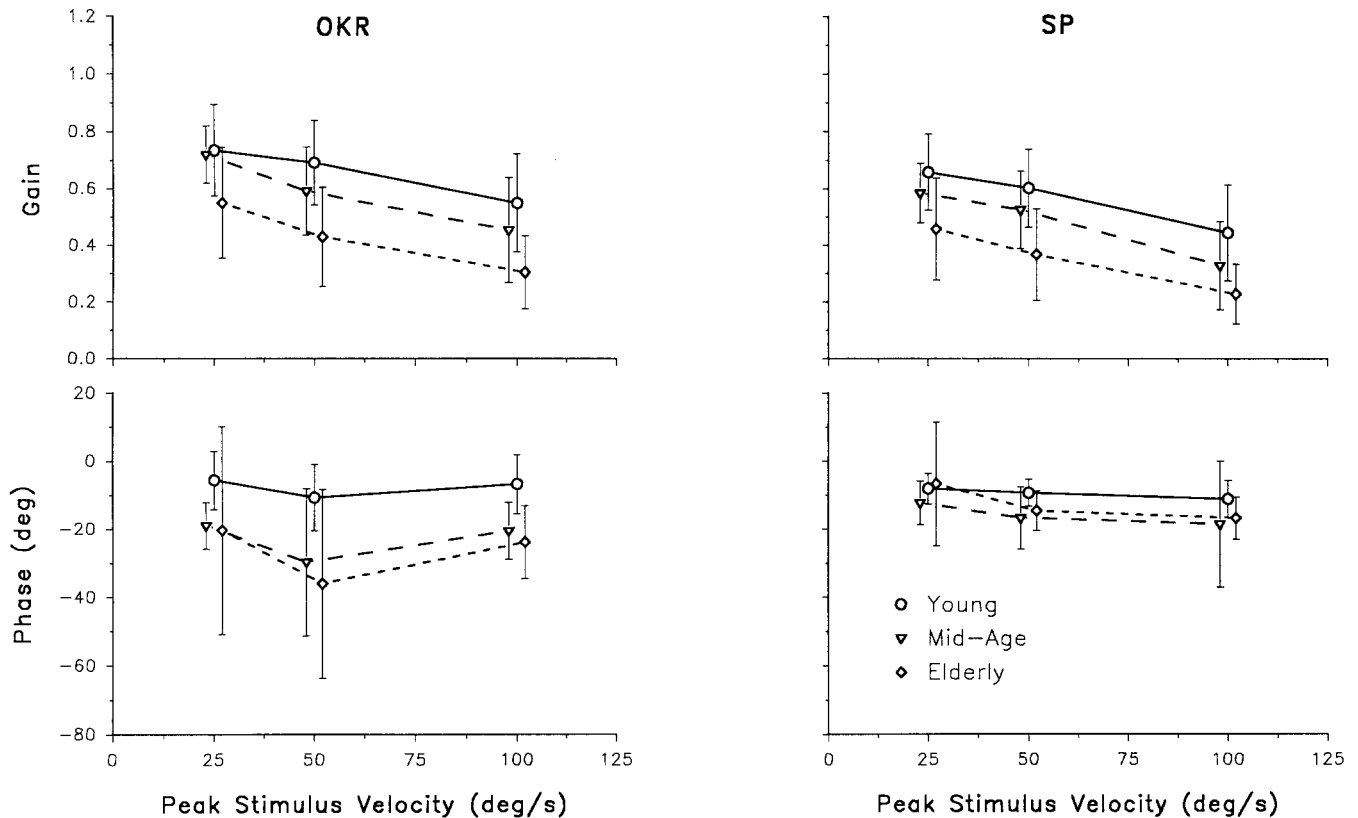


Fig. 4 Mean OKR and SP gain and phase as a function of peak stimulus velocity for all age groups

Relationship between SP and OKR responses

SP and OKR response characteristics are strikingly similar in quality. Many investigators have presumed that the two reflect the same system, at least at higher frequencies. This study provides a direct comparison between SP and OKR responses in the same subjects, under the same conditions, and across ages. One useful comparison between the two was obtained by plotting SP against OKR gain and phase for each subject, regardless of age. The analysis was restricted to data from 0.25-, 1-, and 2.5-Hz trials. The result appears in Fig. 5, and reveals roughly linear relationships between SP and OKR parameters. Regression slopes (solid lines in Fig. 5) measure 0.881 for gain ($r^2=0.83$, $P<0.001$), and 0.591 for phase ($r^2=0.84$; $P<0.001$). Note that these slopes are less than one (dotted lines in Fig. 5). This indicates that OKR gain and phase lag both tend to be greater than corresponding SP values, more so at and above 1 Hz than at 0.25 Hz (see Fig. 5). This is surprising because higher gain is ordinarily associated with smaller phase lag.

In Fig. 5, part of the scatter in response parameters at each frequency is due to age-related effects. If the OKR and SP are closely related, then the effects of aging on each type of visual following should also be tightly linked. Any *difference* between OKR and SP perfor-

mance would then remain the same across ages, even as both deteriorate. Any *change* in the differences as a function of aging would, instead, indicate a selective effect of aging on one or the other. To pursue this concern, differences between OKR and SP gain ($d\text{Gain} = \text{OKR gain} - \text{SP gain}$) and phase ($d\text{Phase} = \text{OKR phase} - \text{SP phase}$) for each subject were computed. Since plots of the differences showed an apparent linear progression with aging, the differences were subjected to linear regression against age for the same frequencies as in Fig. 5. The results proved consistent for phase only. At all three frequencies, OKR phase lag increases progressively more than that of SP as age increases (for regression slopes; $P<0.001$ at 0.25 and 1.0 Hz, and $P<0.01$ at 2.5 Hz).

Another interesting comparison between SP and OKR responses is obtained by plotting response characteristics as a function of peak target acceleration rather than frequency, since it has been reported (Lisberger et al. 1981a; Baloh et al. 1988) that SP is more closely governed by target acceleration than either velocity or position. Figure 6 shows the results. The similarities between the SP and OKR plots, including age-dependent effects, are clear. Note the two sets of outlying points at 251 and 393°/s/s peak acceleration in each plot. These points represent 4- and 2.5-Hz trials, respectively. Ignoring the points from 2.5 and 4 Hz trials, the plots of Fig. 6 resemble data from previous studies (Lisberger et al. 1981a; Baloh et al. 1988), which did not test responses at the highest frequencies used here. The 2.5- and 4-Hz points illustrate that SP and OKR gain (re-ac-

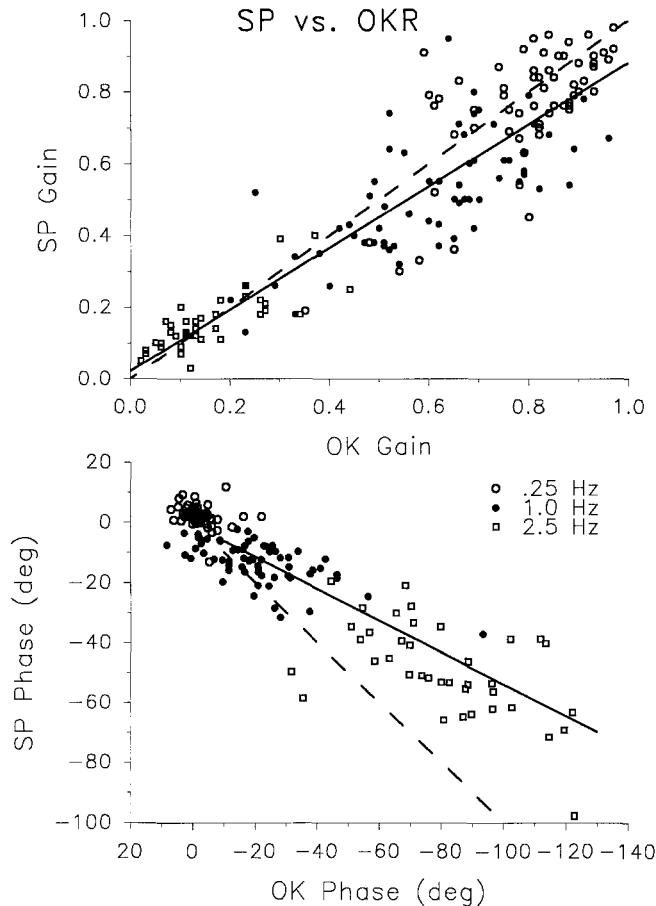


Fig. 5 Comparison between OKR and SP response characteristics. SP gains (*top*) and phases (*bottom*) for each individual are plotted against corresponding OKR gains and phases at three frequencies (see key). The scatter reflects both frequency- and age-dependent influences on response properties. *Solid lines* represent linear regressions, which are highly significant for both gain and phase. *Dotted lines* represent equality between SP and OKR response parameters

celeration) are not independent of frequency, and that frequency-dependent saturation plays an important role above 1 Hz. Phase plots show a similar phenomenon and support the same conclusion.

Latency of visual following

More recent experiments in this study included the analysis of a pre-stimulus period just prior to the onset of oscillation. This facilitated a latency measure, taken as the delay from the onset of OK field oscillation to the initial ocular response. Latency was derived by eye from digital eye velocity records, with a resolution of 5 ms (200 Hz sampling rate). The clearest results were obtained from 2.5-Hz trials. The delay between the start of stimulus oscillation and the initiation of the ocular response is clearly seen in Fig. 1.

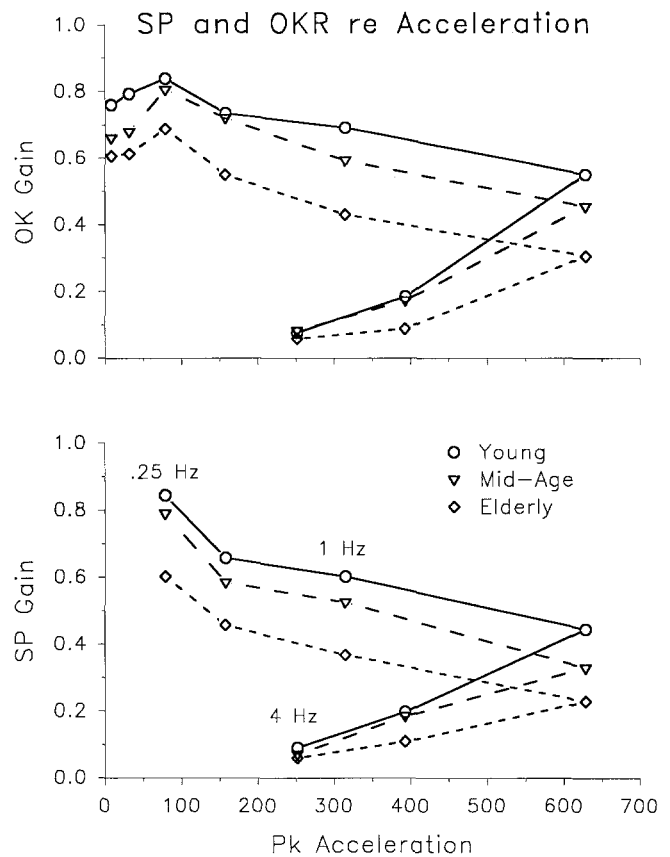


Fig. 6 Mean OKR (*top*) and SP (*bottom*) gains are plotted as a function of peak stimulus acceleration, instead of peak velocity, for all frequencies (0.025–4 Hz) and amplitudes (25–100°/s) in the study, and for each age group (see key). Note the qualitative similarity between the plots, reflecting a similarity in OKR and SP performance across ages. Also note the folding of each plot as frequency (labelled in *bottom traces*) exceeds 1 Hz, but where acceleration declines. Data at 4 and 2.5 Hz, corresponding to 251 and 393°/s/s, respectively, do not fit on the same curve as at lower frequencies

Responses from 32 subjects (nine young, 11 middle-aged, and 12 elderly) were analyzed. The mean latency over all ages measured 167 ± 21 ms. Interestingly, a regression of OKR response latency on age revealed no significant trend. This is in contrast to previous observations in which SP stimuli were used (Sharpe and Sylvester 1978), as opposed to an OK field.

Perhaps the difference in visual stimulus is important in determining the response latency. One relevant concern is that a latency of 167 msec for OK following is longer than that reported for SP, which typically measures 100–150 ms (Tychsen and Lisberger 1986; Carl and Gellman 1987). To determine whether latencies for the initiation of SP and OKR responses differ, direct comparisons were obtained in 15 subjects. SP latency averaged 130 ms, which is indeed shorter than that of the OKR (174 ms in this subgroup), yielding a paired difference of 44 ± 26 ms ($P < 0.001$).

The latency difference could in part reflect a technical distinction between the stimulus onset signals. While the OK stimulus (a velocity servo) begins as a velocity sinusoid, the SP stimulus (a position servo) begins as a position sinusoid (velocity cosine). The greater initial velocity for the SP than for the OK stimulus might shorten the latency for the SP response onset. However, from Carl and Gellman (1987), only target velocities of 5°/s and under affect response latency. For the 25°/s stimulus used here, this factor could only account for at most 13 ms, far short of the 44 ms recorded. This argues in favor of a real difference in response latency between SP and OKR responses.

Circular vection during OK responses

The VOR and the OKR are thought to be intimately related. One potential outcome is the sensation of self rotation that accompanies prolonged OK stimulation, termed "circular vection" (CV) (Dichgans and Brandt 1978). CV is usually studied during constant velocity OK stimuli, in which case the illusion builds gradually over roughly 20 s (Dichgans and Brandt 1978). CV also occurs during low-frequency oscillation of the visual field (Büttner and Henn 1981). In the current experiments, a simple measure of CV was obtained during 0.025-Hz OK stimulation, which was sufficiently prolonged that CV was generated in most subjects. CV was always in the opposite direction to visual image motion, and reversed direction after the image motion reversed. However, CV usually developed after a delay into each half-cycle of stimulation. Thus, CV would typically appear toward the end of the first half-cycle, then disappear as the stimulus reversed, only to reappear in the opposite direction late in the next half-cycle, and so on.

Perhaps a relevant relationship exists between CV and aging. Subjects were asked to identify the moment, if any, when the sensation of self-rotation began, disappeared, and reappeared over the entire 2½ cycle trial. A CV score was assigned, depending upon whether CV was absent (0), barely perceptible at the end of each half cycle (1), or persisted during the last third (2), the middle and last thirds (3), or nearly the entire half-cycle (4). Interestingly, both the likelihood and the intensity of CV increased with aging. This assertion is based upon two measures. First, the proportion of subjects who experienced CV (score > 0) increased from 75% in the young group to 82% in the middle-aged group and to 87% in the elderly. Second, the CV score gradually increased with aging; a regression of CV score on age revealed a significant relationship of +0.024/year ($P < 0.01$), with an intercept at age 0 of 1.1 ($P < 0.05$).

Visual-vestibular interaction across ages

Visual-vestibular interactions were studied using earth-fixed and head-fixed visual stimuli. The different forms of VVI are presented separately below.

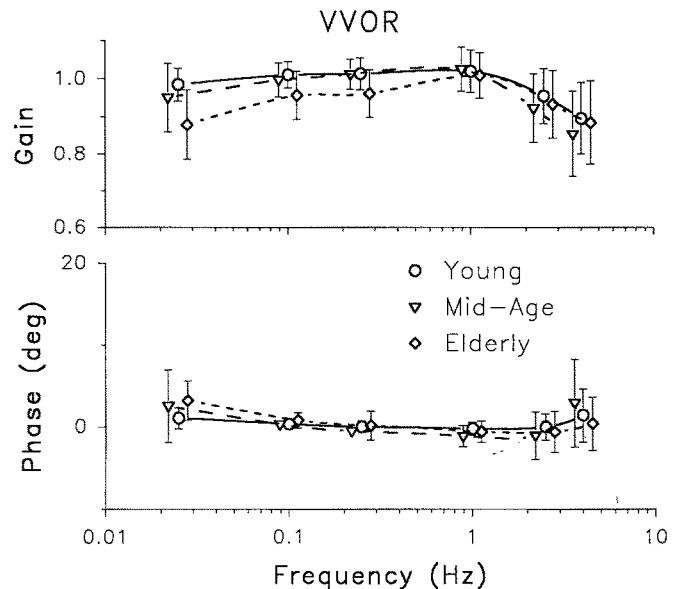


Fig. 7 Mean VVOR gain and phase as a function of frequency for all age groups, plotted as in Fig. 2

Visual enhancement of the VOR (VVOR)

When the head is rotated in the presence of an earth-fixed field, response gain is closer to the ideal of 1.0, while phase is closer to the ideal of 0°, than for the VOR or OKR alone. This is true for all frequencies and all age groups (Fig. 7, Table 1, while Fig. 1 (VVOR) shows the effect in raw form at 2.5 Hz). However, gain for each age group is greatest, and closest to unity, in a modest range of frequencies between 0.1 and 1.0 Hz, and tends to drop off at both frequency extremes. This effect is more pronounced as age increases (Fig. 7). Note that statistically significant age-related differences are restricted to low frequencies (Table 1).

Visual suppression of the VOR (VSVOR)

When the head is rotated together with the OK field (VSVOR-OK) or the SP target (VSVOR-SP), the extent to which visual inputs suppress the VOR depends dramatically upon frequency (Fig. 8). Gain is near zero at 0.25 Hz but rises progressively as frequency increases, reaching values at 4 Hz that are nearly indistinguishable from those of the VOR in darkness. Phase generally displays a lead which is greatest at 1 Hz and which declines at both higher and lower frequencies.

Age-related changes in VSVOR responses were observed (Fig. 8, Table 1), but in a frequency-dependent manner. VSVOR-OK and VSVOR-SP gains tend to rise with increasing age at 0.25 and 1 Hz. However, the opposite occurs at 4 Hz, where gain declines with increasing age (Fig. 8). Corresponding phases, though more variable, typically show the smallest leads in the elderly group across all frequencies. Closer scrutiny across age

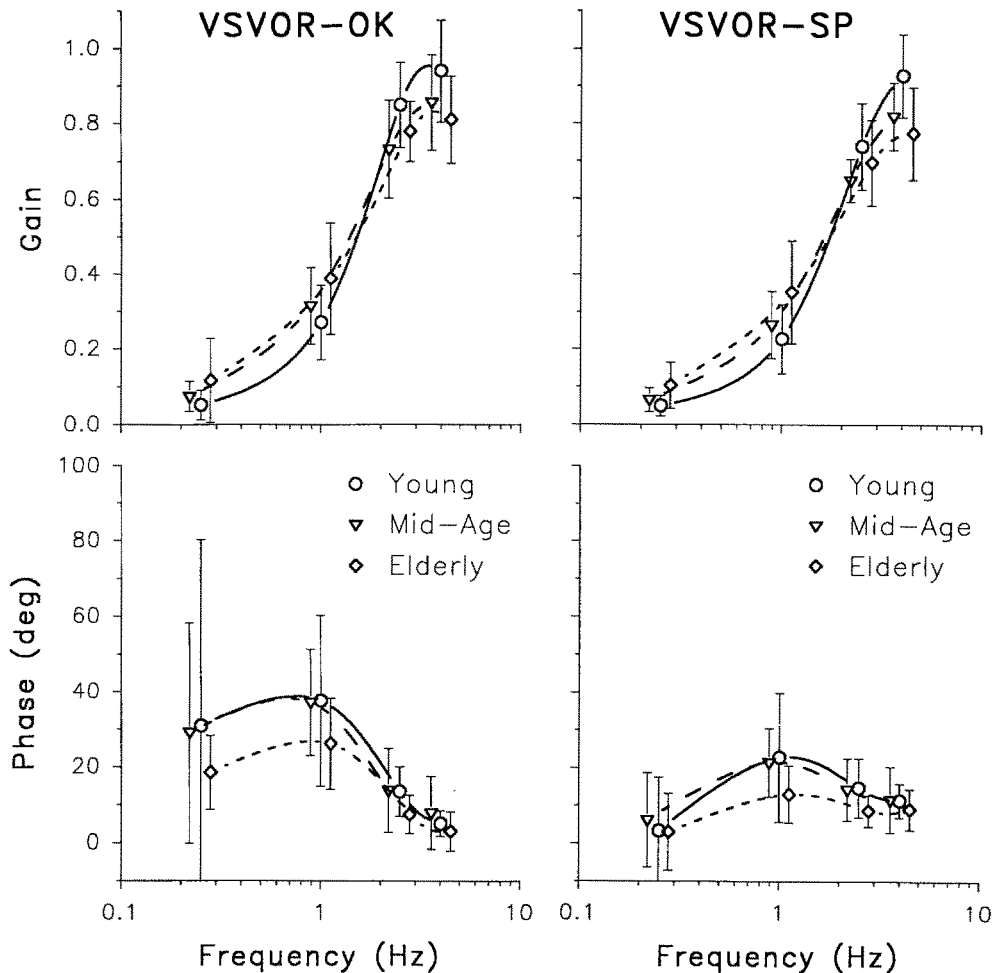


Fig. 8 Mean VSVOR-OK and VSVOR-SP gain and phase as a function of frequency for all age groups, displayed as in Fig. 2

groups at 4 Hz (Table 1) reveals that gain for both types of VSVOR is actually greater than VOR gain in the young group (though significantly so only for VSVOR-OK; $P < 0.05$, paired t test). However, the gain difference vanishes with advancing age. In contrast, phase lead remains significantly greater for both VSVOR-OK and VSVOR-SP responses than for the VOR ($P < 0.001$) over all age groups.

The qualitative similarity between VSVOR-OK and VSVOR-SP response characteristics presumably reflects similar influences of the OKR and SP on the VOR during combined stimulation. A comparison between the two types of VSVOR might provide another argument relevant to the linkage between SP and the OKR. To sharpen the comparison, differences within subjects between VSVOR-OK and VSVOR-SP for both gain and phase were evaluated at each frequency. The mean differences were pooled across ages. VSVOR-OK gains prove slightly greater than VSVOR-SP gains at all but 0.25 Hz, where gain approaches zero regardless of the form of visual stimulus. Further, phase lead tends to be

greater for VSVOR-OK than for VSVOR-SP at 0.25 and 1 Hz, but shifts with increasing frequency to achieve the opposite at 4 Hz. There are no statistically significant age-dependent trends within the differences by regression analysis. This suggests that responses to the two types of visual suppression paradigms shift in parallel fashion as age increases.

Age-dependent progression of VVI changes

The parcelling of the subject pool into three age groups has been useful for initial presentation of the data and for population statistics. However, to more clearly illustrate the actual progression of age-dependent changes in VVI and its components, the population was sectioned into age decades (Paige 1992). Mean gains at several frequencies for each of the six stimulus conditions were then plotted against mean age across decades. The result is shown in Fig. 9.

Note the relative stability of VOR gain across age decades. This is in contrast to most other traces, in which visual influences on eye movements are involved. The subtle decline in VOR gain with increasing age proved insignificant by regression analysis (over all sub-

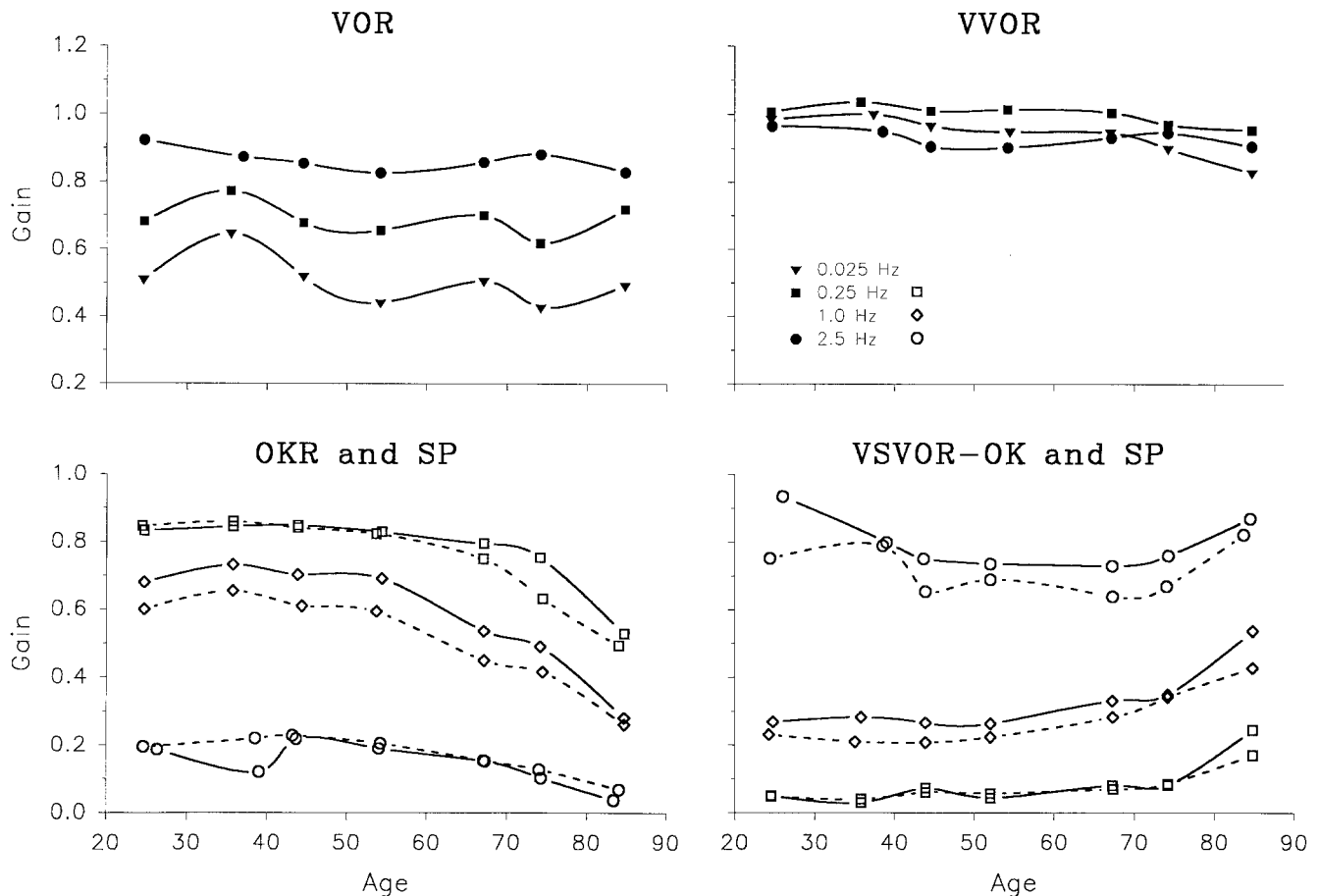


Fig. 9 Mean gains at selected frequencies (see key) are plotted against mean age by age decade for all six stimulus conditions studied. For the bottom two plots, OKR and VSVOR-OK are plotted with *solid lines* while SP and VSVOR-SP are plotted with *dotted lines*. The key distinguishes the frequencies used in the VOR and VVOR plots (*solid symbols*) and in the OKR, SP, and VSVOR plots (*open symbols*)

jects, unparcelled) at each frequency. Results for OKR and SP responses are more dramatic (Fig. 9). In most cases, and at each frequency, gain remains relatively fixed over the first three decades and then declines progressively with increasing age. Again, note the qualitative similarity between OKR and SP traces, and the tendency for OKR gain to remain greater than SP gain over most of the age range, particularly at 1 Hz.

Age-dependent changes in VVI are also illustrated in Fig 9. VVOR gain declines significantly with increasing age at 0.025 Hz ($P < 0.01$ by regression) and 0.25 Hz ($P < 0.05$), but not at 2.5 Hz. VSVOR responses display more complicated behavior. Note the progressive increase in gain with aging at 0.25 and 1 Hz. The curves resemble a mirror image of visually-driven responses (OKR and SP). This is not the case at 2.5 Hz, however. At this higher frequency, gain declines as age increases from the 20s to the 60s, but then rises as age increases into the 80s. This seemingly conflicting finding can in

fact be explained by a linear interaction between visual following and the VOR, as discussed below.

Discussion

Age-dependent changes in the VOR

Age-dependent modifications in the VOR represent modest changes that are most robust and interpretable at low frequencies (i.e., ≤ 0.1 Hz). The phenomenon can be described as a deterioration in low-frequency VOR response characteristics, and consists of an increase in phase lead and decline in gain as age progresses, particularly above 70 years of age (Wall et al. 1984; Peterka et al. 1990a; Paige 1992).

It is perhaps surprising that VOR performance at high frequencies remains remarkably well preserved with aging. However, this may only apply so long as head velocity remains low. This concern has been addressed previously, but only at the modest frequency of 0.25 Hz (Paige and Sargent 1991). As head velocity rises, nonlinearity in the VOR is apparent in the form of an amplitude-dependent deterioration in VOR performance. The nonlinearity is enhanced as age progresses (Paige 1992). The characteristics of the aging reflex, together with anatomical evidence, suggest that aging en-

tails a modest bilateral vestibular deficit, which is presumably counteracted by adaptive mechanisms that help restore VOR function (Paige 1992).

VOR-OKR interaction and velocity storage

Velocity storage and aging

A potential mechanism for the low-frequency VOR deterioration that occurs with aging is a decline in velocity storage (Paige 1992), a central process thought to be shared by the VOR and OKR, and involving the vestibular nuclei (Raphan et al. 1979; Waespe and Henn 1985). Velocity storage is postulated to: (a) enhance low frequency VOR performance relative to that predicted on the basis of its peripheral afferent input; and, (b) provide the substrate for optokinetic after-nystagmus (OKAN), a persistent response in darkness that follows prolonged OK stimulation. Velocity storage in humans is limited to eye velocities up to only 10–20°/s, at which point the process saturates (Segal and Liben 1985; Lafortune et al. 1986; Tijssen et al. 1989; DiZio and Lackner 1990). The best evidence that velocity storage declines with aging is from Simmons and Büttner (1985): the maximum OKAN velocity (saturation point of velocity storage) demonstrably declines with advancing age.

Why would velocity storage decline with increasing age? A reasonable speculation is that velocity storage requires normal peripheral vestibular function, and aging entails a partial bilateral loss of vestibular input (Peterka et al. 1990a; Paige 1992). This notion is based on lesion studies, in which destructive vestibular pathology is associated with a decrease in velocity storage (Raphan and Cohen 1985; Fetter and Zee 1988; Paige 1989). The phenomenon is asymmetric in unilateral loss (Ireland and Jell 1982; Fetter and Zee 1988), but is symmetric and profound in severe bilateral loss (Zee et al. 1976; Ireland and Jell 1982; Raphan and Cohen 1985; Waespe and Wolfensberger 1985). The effect seems related to the degree of vestibular destruction, since humans with partial vestibular loss also show residual OKAN (Zee et al. 1976; Ireland and Jell 1982).

Circular vection during OKR responses

An interesting form of visual-vestibular interaction is the sensation of self-rotation (circular vection, or CV) that accompanies prolonged OK stimulation (Dichgans and Brandt 1978). CV may be an outcome of velocity storage (Dichgans and Brandt 1978). In support, bilateral vestibular loss abolishes or severely impairs CV in rough association with impairment of OKAN (Zee et al. 1976).

Our results seem to contradict the notion that CV is derived from a velocity storage signal shared by the VOR and OKR. This is because of the surprising finding

that CV is *enhanced* with aging, in contrast to the apparent deterioration of velocity storage due to age-related vestibular loss. This does not necessarily imply that vection is derived from a signal that is independent of the vestibular nuclei. Little is known about visual influences on neurons within the vestibular nuclei after vestibular loss, when velocity storage seems compromised. Perhaps visual influences – independent of velocity storage – are enhanced in the vestibular nuclei, so as to adaptively recover self-motion cues by visual means in the absence of adequate vestibular input. This might also explain the robust and enhanced roll-vection observed in some bilaterally (and perhaps partially) vestibulopathic subjects (Cheung et al. 1989). It may be relevant that VOR and OKAN dynamics are dissociated following canal plugging, suggesting in a more direct fashion that velocity storage entails a more complex relationship between the VOR and OKR than is commonly believed (Paige 1983b). As an alternative to a brainstem substrate, CV may result from interactions between cortical visual and vestibular areas (Straube and Brandt 1987; Heide et al. 1990; Büttner and Henn 1981). Following reduction of vestibular input (e.g., with aging), visual influences could adaptively recovery self-motion cues just as readily in cortex as in the brainstem.

Smooth pursuit and optokinetic responses

Smooth pursuit of a punctate target (SP) and visual following of a large-field scene (OKR) were studied in the same individuals. This allowed a direct comparison between the SP and OKR, as well as an assessment of age-dependent changes in their performance. The comparison was motivated by an ongoing controversy as to whether SP and the OKR are the same, or whether SP is inherently contained within OKR responses. Although previous studies of SP (Spooner et al. 1980; Zackon and Sharpe 1987; Larsby et al. 1988) and OKR (Simons and Büttner 1985; Peterka et al. 1990b) have reported age-dependent deteriorations, differences in methods and subjects cloud direct comparisons between the two types of visual following.

The OKR is generally thought to include two components, one fast and the other slow (Cohen et al. 1977; Lisberger et al. 1981b; Robinson 1981; Paige 1983a). At frequencies below around 0.1 Hz, the fast and slow components operate in concert, while at frequencies above 0.1 Hz, the slow component drops out, leaving the fast component in isolation. The fast component of the OKR has been likened to SP, and may be the same (Lisberger et al. 1981b; Robinson 1981; Paige 1983a; Yasui and Young 1984). The close relationship between SP and OKR response properties observed here at and above 0.25 Hz are largely in keeping with previous reports (Lisberger et al. 1981a; Paige 1983a; Yee et al. 1983; Collewijn and Tamminga 1984; Yasui and Young 1984; Boyle et al. 1985; Van den Berg and Collewijn 1986; Wyatt and Pola 1988). In addition, this study

demonstrates that the linkage between SP and OKR response properties largely holds across ages even as performance deteriorates in both.

The close similarity between SP and OKR behavior does not mean that the response characteristics of the two types of visual following are indistinguishable, only that they are closely related and fluctuate together. If SP and the fast-component of the OKR are the same, then how can the subtle but measurable quantitative differences between them be explained? In general, gain and phase lag of OKR tend to be greater than that of SP across stimulus frequencies and amplitudes, as well as across ages. A simple explanation for the gain differences is that the OK stimulus provides a more powerful stimulus to the SP system than does a small punctate target (Van den Berg and Collewijn 1986). The OK stimulus activates a larger area of retina, and therefore a greater number of responsive visual elements, than the SP target. The OK stimulus might then generate a greater response. However, this explanation does not account for the greater phase lag in OKR than in SP responses. The phase difference might be explained by a summation phenomenon in which multiple widespread visual elements, each with different speeds and amplitudes of response, are simultaneously activated by the large-field stimulus. The fastest and most robust elements would be driven by peri-foveal inputs, and progressively more sluggish and weaker elements by increasingly peripheral retinal inputs. The summed signal would result in a larger but more lagged response than that driven by central retinal elements alone. This hypothesis receives support from Tychsen and Lisberger (1986), who studied the initiation of human SP in response to focal stimulation of the retina at different eccentricities. Stimulation within several degrees of the fovea generated the greatest and fastest responses, while more eccentric stimuli produced responses with smaller amplitudes and accelerations, but longer latencies.

A similar explanation might account for the difference in latency between OKR and SP initiation. The relative delay between the initiation of SP and OKR responses should be directly related to the phase difference. The relative phase lag (θ) between SP and OKR due to a latency difference (L) can be readily calculated for each frequency (f) according to the relation, $\theta = 360 \cdot L \cdot f$. For the subpopulation in which OKR and SP latencies were measured, the latency difference averages 44 ms. This translates into a phase difference of -4° , -16° , -40° , and -63° for frequencies of 0.25, 1, 2.5, and 4 Hz, respectively. These differences are somewhat greater than those measured in young subjects, but are quite close to those in middle-aged and elderly subjects.

One caveat in this line of reasoning is that predictive effects during sinusoidal visual following may differ between SP and OKR responses. If so, then the relationship between response latency and phase would explain only a part of the phase difference. Another concern is whether predictive capabilities might themselves deteri-

orate with aging, since this might account for a general age-dependent deterioration in visual following. In our experiments, prediction in SP and the OKR are evidenced by the absence of phase lag, and even the presence of small phase leads, at modest frequencies, as reported by others (Robinson 1981; Bahill and McDonald 1983; Wyatt and Pola 1988). Although we cannot directly assess whether predictive influences are affected by aging or are the same in SP and OKR responses, relevant data are provided by Larsby et al. (1988). They studied SP during both periodic and non-periodic tracking over a broad age range, and found no evidence that prediction was lacking in the elderly. An alternative factor might be an age-dependent deterioration in attention to visual targets or visuo-motor tasks.

Visual-vestibular interactions

Overview of VVOR and VSVOR responses

Most natural head movements occur in the presence of vision, so that eye movements are driven by a combination of vestibular and visual signals. When the visual world is earth-fixed, vestibular and visual signals combine synergistically to maintain fixation (VVOR), and eye movements are generally more accurate (gain and phase closer to 1.0 and 0° , respectively) than when either vestibular or visual signals operate alone. This was observed here and in previous studies using a variety of primate species (Barr et al. 1976; Paige 1983a; Barnes 1988; Jell et al. 1988; Kasteel-van Linge and Maas 1990). In addition, this investigation provides evidence for age-dependent deterioration. Mean VVOR gain is lower in the elderly group than in the young group at every frequency. This is expected, since visual following displays a robust age-dependent decline in performance, and therefore would not enhance the VOR as well in the elderly as in younger subjects. Further, VOR performance declines in the elderly at the lowest frequencies, and this combines with the deficit in visual following to compound a decline in VVOR responses at these frequencies.

When the head and visual target move in tandem, visual following serves to suppress the VOR (VSVOR), and the subject appears to track with the head while the eyes remain fixed in their orbits. However, this phenomenon only applies to low and modest frequencies of head movement. Above around 1 Hz, visual suppression fails and cannot overcome the VOR. By around 4 Hz, no significant visual suppression of the VOR occurs, regardless of age or whether the visual stimulus is a small target (VSVOR-SP) or a large field (VSVOR-OK). Indeed, VSVOR gain at 4 Hz is typically slightly greater than that of the VOR (and even VVOR, see Table 1). Similar VSVOR characteristics have been repeatedly demonstrated (Paige 1983a; Furst et al. 1987; Barnes 1988; Jell et al. 1988; Kasteel-van Linge and Maas 1990). The reasons for this seemingly anomalous behavior are discussed along with a model.

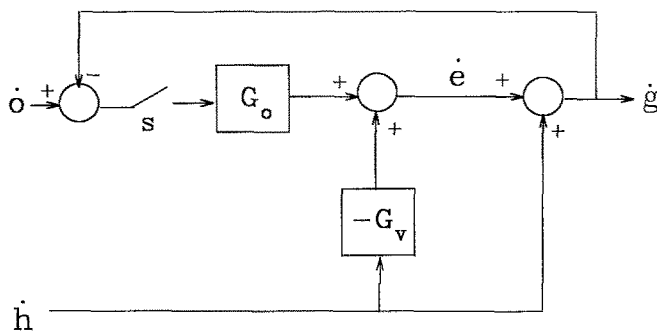


Fig. 10 Simplified linear model of visual-vestibular interactions. Head ($\dot{h}$) and visual image ($\dot{o}$) velocities in space serve as inputs. Each is associated with a dynamic element, G_v and G_o , respectively. G_v is negated to correspond with the compensatory nature of the VOR. The "retinal" visual input to G_o , retinal slip, is derived as the difference in velocity between the image in space and the eye in space (gaze, $\dot{g}$). The relevant output is eye velocity in the head ($\dot{e}$). The switch in the visual path (s) is closed only when a trackable image is available. The following relations hold: $*VOR = -\dot{e}/\dot{h} = G_v/(1 + G_o)$; $*OKR = \dot{e}/\dot{h} = G_o/(1 + G_o)$; $*VSVOR = -\dot{e}/\dot{h} = G_v/(1 + G_o + G_v)$; $*VVOR = -\dot{e}/\dot{h} = (G_v + G_o)/(1 + G_o + G_v)$. Eqs. 1 and 2 are the same as the latter two relations. These would still hold true even after including additional dynamic elements and internal feedback pathways that embellish more complex models

Model of visual-vestibular interactions

A persistent and controversial issue is whether the VOR and visual following mechanisms interact linearly to generate ocular responses under the various conditions of VVI. The question can be addressed by comparing actual VVI responses with predictions derived from VOR and visual (OKR and SP) responses in isolation. However, rather than present a comprehensive control systems model of VVI and its components, our purposes are best served by a simpler approach that focuses directly on vectorial interactions between vestibular and visual signals driving eye movements. The model of Fig. 10 serves our purposes well, and is intended only to formalize the desired relationships between the VOR, visual following, and VVI. The following vectorial equations are derived from the model, but do not depend upon its simplified structure (see Paige 1983a):

$$*VSVOR = VOR(1 - OKR) \quad (1)$$

$$*VVOR = OKR + VOR(1 - OKR) = VOR + OKR(1 - VOR) \quad (2)$$

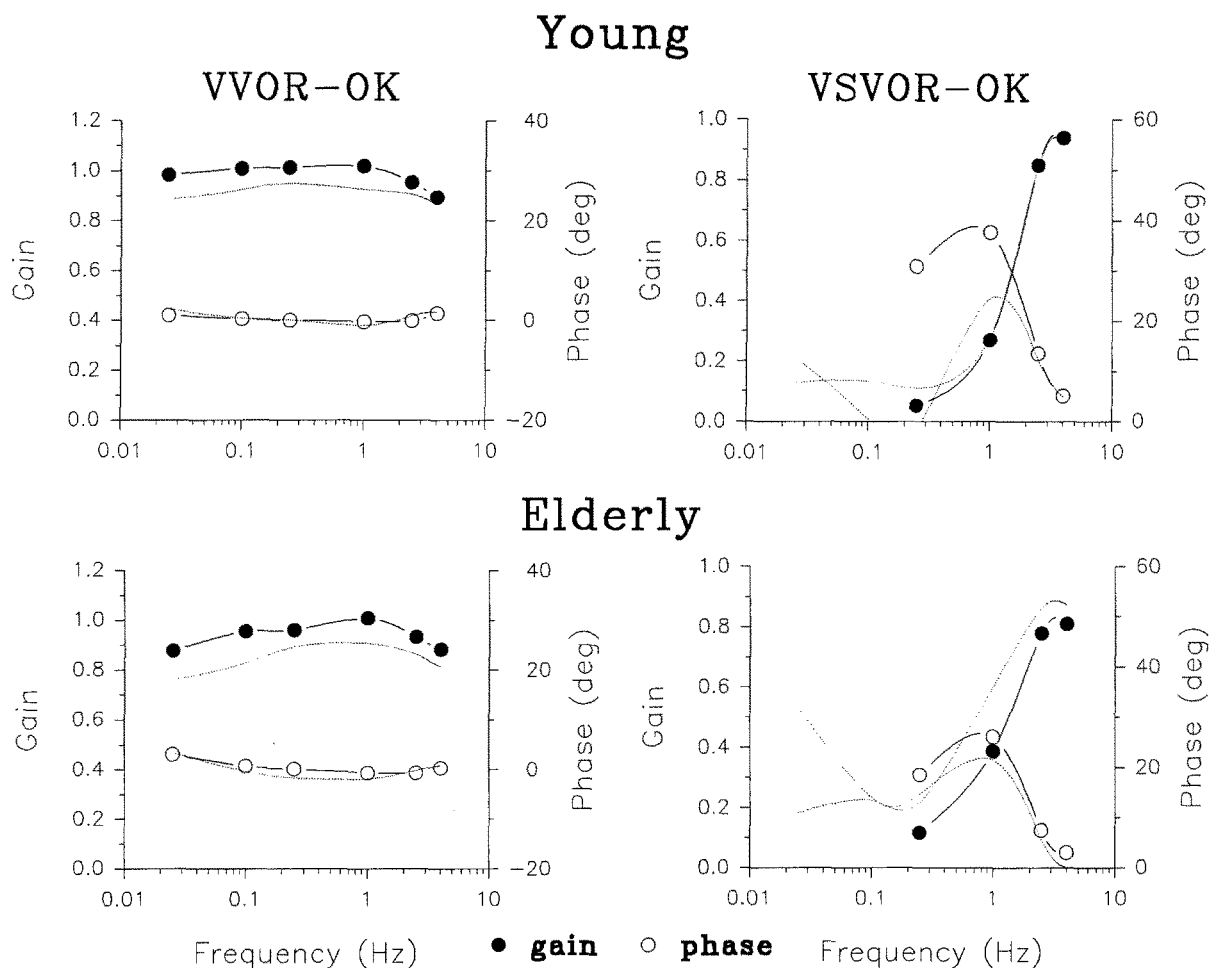


Fig. 11 Mean VVOR-OK (left) and VSVOR-OK (right) gains (solid dots) and phases (open circles) are plotted against frequency for the Young (top graphs) and elderly (bottom graphs) groups,

together with values calculated from a linear model of VVI according to Eqs. 1 and 2. VOR and OKR data from Table 1 were used in these calculations

where the “*” denotes a calculated rather than a measured value.

These equations were used to calculate gains and phases of *VVOR and *VSVOR-OK from actual VOR and OKR response parameters at each frequency. By substituting SP for OKR in Eqs. 1 and 2, *VSVOR-SP characteristics were similarly derived. The results for the young and elderly are shown in Fig. 11 as fine lines associated with actual VVOR and VSVOR measurements. Although there is a qualitative similarity between VVI responses and those predicted by Eqs. 1 and 2, quantitative discrepancies are visible. For example, actual VVOR gains are typically greater than predicted, while VSVOR gains are usually lower. These discrepancies decline as frequency rises, and are nearly gone at 4 Hz.

Despite such differences, the model helps explain some potentially confounding elements of VVI responses. First, VSVOR gain at 4 Hz *declines* with increasing age, in contrast to the VOR alone (Fig. 8). The phenomenon is linked with another apparent anomaly – VSVOR gain is greater than VOR gain at 4 Hz. The reason for both observations is that visual following at this high frequency actually enhances rather than suppresses the VOR under VSVOR conditions. Note that OKR and SP phase lag at 4 Hz is greater than -90° . Any gain available then *adds* slightly to the VOR. The age-dependent decline in VSVOR occurs because OKR and SP responses are most robust in the young, and so the gain increment is also most robust in this group, adding less to the VOR as age progresses. Similar reasoning can explain the age-dependent increase in VSVOR gains at the lowest frequencies. Since OKR and SP responses are robust, and phase lags are near zero, head-fixed visual inputs result in strong VOR suppression. As visual following declines with increasing age, the VOR is less readily suppressed, yielding larger VSVOR gains.

A second and related nuance of VSVOR responses occurs at 2.5 Hz, in which gain declines with increasing age through the first four to five decades but then rises (Fig. 9, also seen in Fig. 8 and Table 1, where gain at 2.5 Hz is greatest in the young and lowest in the middle-aged, not the elderly, group). The VSVOR-SP data are replotted in Fig. 12 along with model (*VSVOR-SP) results to illustrate that this complex behavior is replicated by a linear VVI model. In both actual (top) and model (bottom) diagrams, gain at 2.5 Hz is lowest in the middle-aged group. Although the largest gain varies (in the young for the actual data and in the elderly for the model), a variety of patterns can be achieved within a narrow frequency range around 2.5 Hz; the important point is that the model pattern of change resembles the actual one.

Another initially perplexing concern is that VSVOR gains at 1 Hz and above are typically greater with the OK scene than with the SP target (see Fig. 9). This is odd because OKR gains are generally greater than SP gains, and the intuitive expectation is that suppression would be better with the more robust OKR. The reason for the

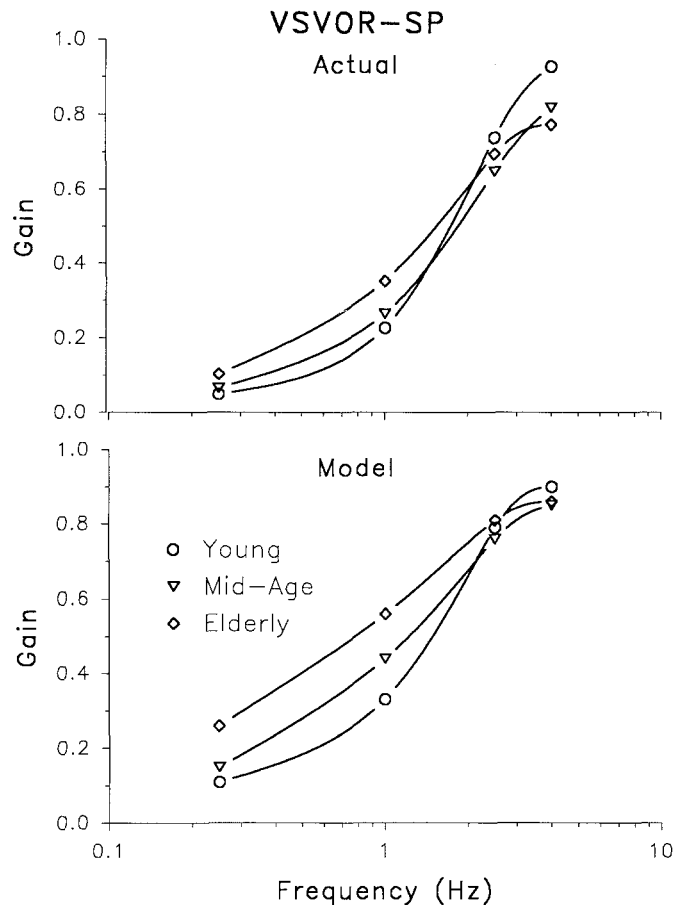


Fig. 12 Mean actual (top) and model (bottom) VSVOR-SP gain as a function of frequency for all age groups, displayed as in Fig. 8, but without x-axis offsets, for each age group. Note the qualitative similarity between the two graphs and the cross-overs in the plots between age groups in the vicinity of 2.5 Hz

result is that OKR phase lags are greater than those of SP, and vectorial interactions with the VOR are affected by both gain and phase. Using Eqs. 1 and 2, the same pattern is revealed by the linear model.

The qualitative similarities between model and actual VVI response characteristics cannot hide their quantitative differences. These differences would seem to argue that VVI is not a linear combination of vestibular and visual influences on eye movements. However, the issue is confounded by an important feature of the human VOR; that is, it can be modulated by the “context” in which VVI takes place (Barr et al. 1976; Melvill Jones et al. 1984). Thus, simply imagining that a target in darkness is head- or earth-fixed results in a VOR decrement or enhancement, respectively, relative to the default performance of the VOR (with no “fixational” task). However, the influence of context decreases as frequency increases (Barr et al. 1976; Furst et al. 1987; Barnes 1988; Kasteel-van Linge and Maas 1990). In this regard, the phenomenon behaves with dynamic limitations that resemble SP.

Which “VOR” is being enhanced or suppressed during VVI tasks? Previous assessments of VVI linearity

Model VOR

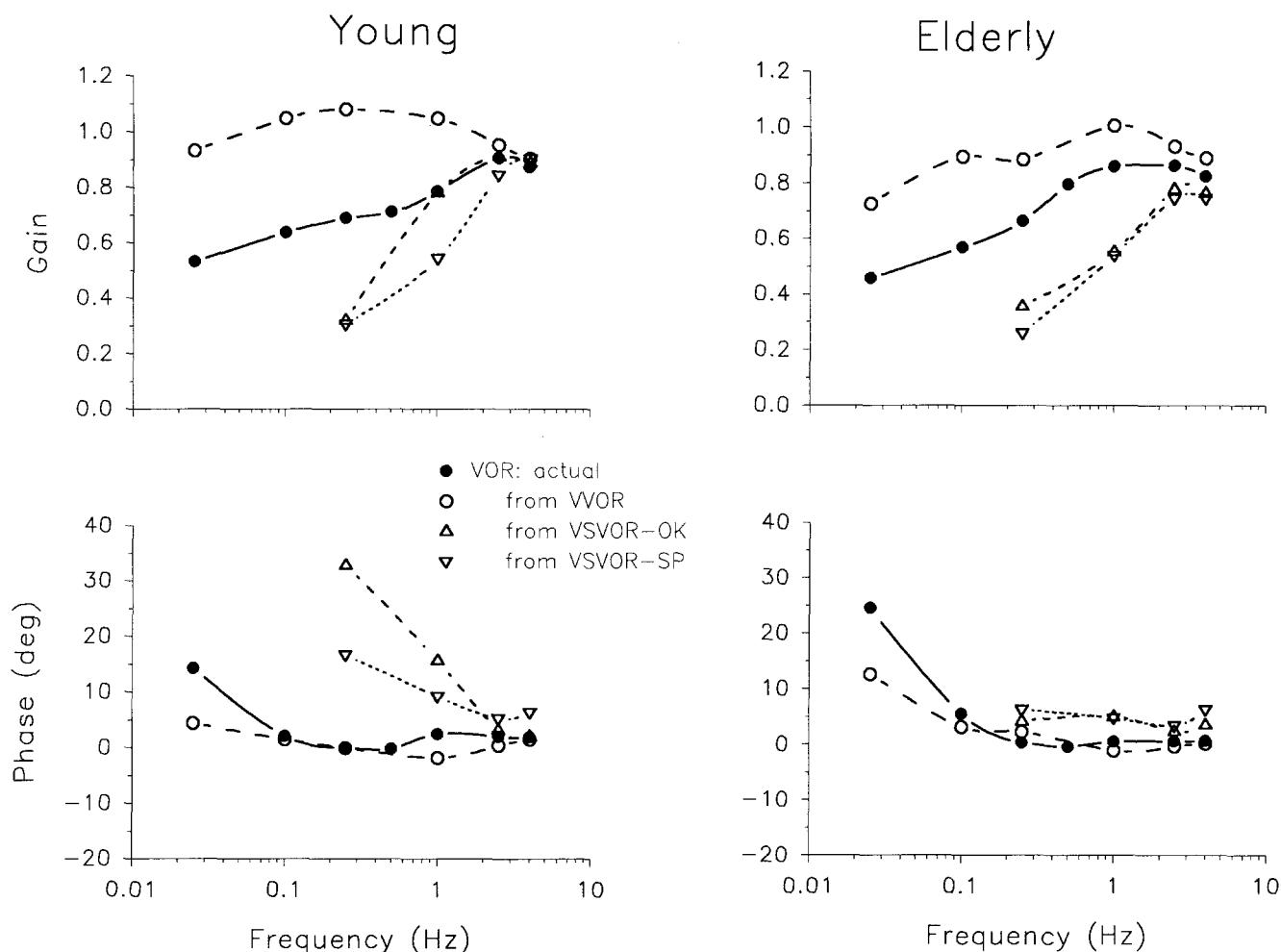


Fig. 13 Actual mean VOR gains and phases (solid circles) are plotted against frequency for the young (left) and elderly (right) groups, together with values calculated from a linear model of VVI according to Eqs. 3 and 4, and using VVI data from Table 1 (see key for origin of data)

have assumed that the VOR is a single entity (Paige 1983a; McKinley and Peterson 1985), as measured in darkness but without a set context. This may be erroneous, because the VOR that is being enhanced under VVOR conditions is presumably of greater gain than the default VOR, taking into account contextual influences on the *VOR component* of VVI (Waterston and Barnes 1992). Similarly, the VOR that is being suppressed under VSVOR conditions is likely of lower gain than the default. The VOR reported here was recorded after subjects were instructed to imagine an earth-fixed field, but while also indulged in naming tasks to maintain vigilance. The recordings most likely represent performance close to the default. This contention is supported by the routine appearance of nystagmus, rather than smooth tracking, in response to VOR trials at 0.5 Hz and below.

Although a direct measure of the context-dependent VOR component of VVI is lacking, estimates can be derived after modifying Eqs. 1 and 2 to calculate model *VOR from recorded VVI data:

$$*VOR_{vs} = VSVOR/(1 - OKR) \quad (3)$$

$$*VOR_{vv} = (VVOR - OKR)/(1 - OKR) \quad (4)$$

where the subscripts indicate values based on visual suppression (vs) or enhancement (vv) paradigms. Figure 13 displays the actual VOR data together with model (*VOR) results for the young and elderly groups (both SP and OK visual inputs were used, after appropriate substitutions in Eqs. 3 and 4).

As expected, the model *VOR values are quite different from the measured VOR in several ways. First, the differences are frequency dependent. Actual and synthetic *VORs are nearly the same at 4 Hz, in keeping with the limited influences of both vision and context on vestibular-driven eye movements at this high frequency. As frequency declines, differences grow. Second, *VOR_{vv} displays a higher gain and lower phase lead than the VOR at all frequencies except 4 Hz, while

*VOR_{vs} shows the opposite pattern. These trends qualitatively resemble the experimental data reported by others (Barr et al. 1976; McKinley and Peterson 1985; Furst et al. 1987; Jell et al. 1988; Barnes 1988). However, in the young group, the differences between model and actual VORs seem larger than these studies suggest. Perhaps VVI includes more complex non-visual influences in addition to context-dependent modifications of the VOR component. Evidence has been presented in monkeys (Lisberger 1990; Cullen et al. 1991); changes in eye movement responses after sudden head perturbations during VVI occur sooner than known visual or contextual mechanisms would allow. However, comparable experiments in humans (Vercher and Gauthier 1991) failed to observe these early effects, and instead suggest that the human VOR remains unmodifiable by vision and context for roughly 150–200 ms. This result was confirmed here by comparing the VOR and VSVOR ($n=32$) during the 25–75 ms interval after the initiation of ocular responses to rotation at 2.5 Hz; VOR gain averaged 0.81 ± 0.15 , while VSVOR-OK gain averaged 0.76 ± 0.17 ($P > 0.05$; paired t test). The discrepancy between VVI behavior and the model remains mysterious. Interestingly, the issue is of concern only in the young group, since model *VORs seem quite realistic in the elderly. It may be that contextual influences are most robust in the young and decline with aging.

In conclusion, eye movements were studied in response to isolated vestibular and visual stimuli, as well as during specific combinations, in order to assess synergistic and antagonistic visual-vestibular interactions. A novel approach was to study how the various subsystems driving smooth eye movements and their interactions change with senescence. Aging entails a progressive and quantifiable deterioration in the ability to control eye movements driven by both vestibular and visual inputs. These systems are important for ensuring the maintenance of gaze on visual targets, and therefore retinal image stability, during natural behavior. Failure of vestibular and visual control of gaze presumably results in disintegration of visual stability and orientation in space. These factors may in part underlie the progressive deterioration in equilibrium and postural control (Radebaugh et al. 1985) that also accompanies the aging process.

Acknowledgements The author thanks D. Fishel and C. Ha for their technical assistance, and Dr. W.M. King for review of the manuscript. Dr. C.R. Fox participated in some of the experiments. This research was supported by NIH grant RO1-AG06442. Many of the elderly subjects were obtained through a program project on falls and hip fractures, supported by NIH grant PO1-AG06815 (Dr. W. Peck, Director). All data for this study were recorded at the Washington University School of Medicine, while the author was on the faculty at that institution. All experiments were performed after approval by the Human Studies Committee of the Washington University School of Medicine.

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